



# The unbeatable random walk in exchange rate forecasting: Reality or myth?



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## ARTICLE INFO

### Article history:

Received 7 May 2013

Accepted 4 March 2014

Available online 22 March 2014

### JEL classification:

F31

F37

C53

### Keywords:

Forecasting

Random walk

Exchange rate models

Direction accuracy

Monetary model

## ABSTRACT

It is demonstrated that the conventional monetary model of exchange rates can (irrespective of the specification, estimation method or the forecasting horizon) outperform the random walk in out-of-sample forecasting if forecasting power is measured by direction accuracy and profitability. Claims of outperforming the random walk in terms of the root mean square error are false because they are typically based on the introduction of dynamics, hence a random walk component, commonly without testing for the statistical significance of the difference between root mean square errors. And even if proper hypothesis testing reveals that a dynamic model outperforms the random walk, this amounts to beating the random walk by a random walk with the help of some explanatory variables. The failure of conventional macroeconomic models to outperform the random walk in terms of the root mean square error should be expected rather than considered to be a puzzle.

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## 1. Introduction

Since the publication of the highly-cited paper of Meese and Rogoff (1983), it has become something like an undisputable fact of life that conventional exchange rate determination models cannot outperform the naïve random walk model in out-of-sample forecasting. This view is still widely accepted to the extent that it is typically argued that the Meese–Rogoff results, which are “yet to be overturned”, constitute a puzzle. For example, Abhyankar et al. (2005) describe as a “major puzzle in international finance” the inability of models based on monetary fundamentals to outperform the random walk. Evans and Lyons (2005) suggest that the Meese–Rogoff finding “has proven robust over the decades”. In another study they describe the finding as “the most researched puzzle in macroeconomics” (Evans and Lyons, 2004). Furthermore, Frankel and Rose (1995) argue that the negative results have had a “pessimistic effect” on the field of exchange rate modeling in particular and international finance in general. Likewise, Bacchetta and van Wincoop (2006) point out that the poor explanatory power of existing exchange rate models is most likely the major weakness of international macroeconomics. Empirical studies of exchange rate models typically corroborate the Meese and Rogoff results.

Several reasons have been put forward for the failure of exchange rate models to outperform the random walk. In their original paper, Meese and Rogoff (1983) attributed the failure to simultaneous equations bias, sampling errors, stochastic movements in the true underlying parameters, misspecification and nonlinearities (hence all of their explanations are

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related to the underlying econometrics). While Voss and Willard (2009) do not consider the forecasting accuracy of exchange rate models, they emphasize the point that monetary policy innovations have asymmetric effects on the exchange rate, which means that imposing the assumption of symmetry may be yet another reason for the failure to outperform the random walk.<sup>1</sup> Meese (1990) adds other explanations such as improper modeling of expectations and over-reliance on the representative agent paradigm. By referring to this paradigm, Meese seems to be questioning the theoretical pillars of conventional exchange rate models.

Contrary to what Meese and Rogoff found, some economists claim that it is possible to outperform the random walk (using the root mean square error and similar metrics as criteria) in the medium or long run (for example, Mark, 1995; Chinn and Meese, 1995; MacDonald, 1999; Mark and Sul, 2001). However, Berben and van Dijk (1998) and Berkowitz and Giorganni (2001) criticize these studies, particularly the assumption of a stable cointegrating relation. Furthermore, Cheung et al. (2005) find that a wide range of models are not successful in forecasting exchange rates. Based on the root mean square error and similar metrics, the Meese and Rogoff results cannot be overturned and are still largely perceived to represent a puzzle. In fact we will find out that claims of the ability to outperform the random walk in terms of the root mean square error are based on flawed procedures. Emphasis must be placed on the phrase “root mean square error and similar metrics”.

Two questions arise out of this state of affairs: (i) is failure to outperform the random walk a puzzle that constitutes failure of international monetary economics?; and (ii) is the random walk unbeatable in exchange rate forecasting? This paper addresses both of these questions. In particular, we endeavor to demonstrate that the “unbeatable random walk” is a myth by exploring the attempts that have been made to resolve the puzzle and by conducting some alternative empirical tests. Specifically, we re-examine the puzzle by (i) using measures of direction accuracy and profitability, in addition to the root mean square error, to judge forecasting power; (ii) introducing dynamics; and (iii) using time-varying parameters. To explore the possibility that models are more capable of outperforming the random walk over long forecasting horizons, we generate forecasts over horizons ranging between one month and six months.

We use only one of the models considered by Meese and Rogoff, the flexible-price monetary model, and justify this approach on three grounds.<sup>2</sup> First, in an exercise like this, one has to be selective with respect to model choice as to keep the results manageable. Cheung et al. (2005) argue that “any evaluation of these models must necessarily be selective” because “the universe of empirical models that have been examined over the floating period is enormous”. Even within the class of non-linear models, the choice is “infinite” (Taylor et al., 2001). Second, Moosa and Burns (2013a) have done similar work on the three macroeconomic models estimated by Meese and Rogoff (1983), using data covering the late 1970s and early 1980s. Their results turned out to be consistent across the three models. Likewise, Fullerton et al. (2001) use the monetary model and obtain results that they describe as “fairly weak for both specifications irrespective of the interest rate variable selected”. The same conclusion is reached by Cheung et al. (2005) for a “wide variety of models”. Last, but not least, demonstrating that at least one of the three models used by Meese and Rogoff outperforms the random walk is adequate for overturning their results and busting the myth. This is because they show that all of the three models fail to outperform the random walk.

As a preview, our results show that the random walk cannot be outperformed in terms of the root mean square error, which depends on the magnitude of the error only, irrespective of the forecasting horizon. In no case does hypothesis testing reveal that the static monetary model has a significantly (or even numerically) lower RMSE than that of the random walk. Introducing dynamics may alter this finding, but this procedure is flawed because it boils down to beating the random walk with another random walk (with the help of some explanatory variables). We argue that failure to outperform the random walk in terms of the RMSE is by no means a puzzle and that we should expect nothing but that. It is only in this sense that the results of Meese and Rogoff cannot be overturned. However, when forecasting power is measured in terms of direction accuracy and profitability, it is rather easy to outperform the random walk.

## 2. The benchmark results

### 2.1. Specification and testing

The basic flexible-price monetary model is specified as

$$s_t = \alpha_0 + \alpha_1(m_{a,t} - m_{b,t}) + \alpha_2(y_{a,t} - y_{b,t}) + \alpha_3(i_{a,t} - i_{b,t}) + \varepsilon_t \quad (1)$$

where  $s$  is the log of the exchange rate,  $m$  is the log of the money supply,  $y$  is the log of industrial production,  $i$  is the interest rate, and  $a$  and  $b$  refer to the countries having  $a$  and  $b$  as their currencies, respectively (the exchange rate is measured as the price of one unit of  $b$ —that is,  $a/b$ ). The model is estimated over part of the sample period,  $t = 1, 2, \dots, m$ , then a  $k$ -period-ahead forecast is generated for the point in time  $m + k$ . The forecast log exchange rate is

<sup>1</sup> With respect to the monetary model, asymmetry may mean one of two things: (i) the exchange rate responds differently to positive and negative changes in the explanatory variables or, when an error correction model is used, to positive and negative deviations from the long-run equilibrium condition; or (ii) relaxation of the assumption typically used in the construction of the monetary model about the cross-country equality of the income elasticities and interest semi-elasticities of the demand for money. Voss and Willard (2009) are concerned with symmetry in the first sense.

<sup>2</sup> These are the Frenkel–Bilson, Dornbusch–Frankel and Hooper–Morton models. None of these models in a basic static form turned out to be better than the other two as compared with the random walk.

$$\hat{s}_{m+k} = \hat{\alpha}_0 + \hat{\alpha}_1(m_{a,m+k} - m_{b,m+k}) + \hat{\alpha}_2(y_{a,m+k} - y_{b,m+k}) + \hat{\alpha}_3(i_{a,m+k} - i_{b,m+k}) \quad (2)$$

where  $\hat{\alpha}_i$  is the estimated value of  $\alpha_i$ . Hence the forecast level of the exchange rate is

$$\hat{S}_{m+k} = \exp(\hat{s}_{m+k}) \quad (3)$$

The process is then repeated by estimating the model over the period  $t = 1, 2, \dots, m+1$  to generate a forecast for the point in time  $m+k+1$ , and so on until we get to  $\hat{s}_n$  by estimating the model over the period  $t = 1, 2, \dots, n-k$ , where  $n$  is the total sample size. This process, therefore, involves recursive regression, which is in line with what is recommended by [Marcellino \(2002\)](#) and [Marcellino et al. \(2001\)](#) who make it explicit that their forecasts are generated by using a “fully recursive methodology”. Preference for recursive over rolling estimation may be justified in terms of forecasting efficiency, which refers to the property that a forecast contains all information available at the time of the forecasts ([Nordhaus, 1987](#)). Information is lost in rolling estimation because some observations are excluded from the sample to obtain a constant estimation window.

Once we have corresponding time series for the actual,  $S_t$ , and forecast,  $\hat{S}_t$ , exchange rates for the period  $t = m+k, \dots, n$ , we can calculate measures of forecasting accuracy based on the percentage forecasting error. Since various quantitative measures of forecasting accuracy lead to the same ranking of models, we use the root mean square error, which is calculated as

$$\text{RMSE} = \sqrt{\frac{1}{n-m-k+1} \sum_{t=m+k}^n \left( \frac{\hat{S}_t - S_t}{S_t} \right)^2} \quad (4)$$

To facilitate comparison with the random walk, we also report Theil's inequality coefficient ( $U$ ), which is calculated as the ratio of RMSE of the model to that of the random walk

$$U = \frac{\sqrt{\frac{1}{n-m-k+1} \sum_{t=m+k}^n \left( \frac{\hat{S}_{t+1} - S_{t+1}}{S_{t+1}} \right)^2}}{\sqrt{\frac{1}{n-m-k+1} \sum_{t=m+k}^n \left( \frac{S_t - S_{t+1}}{S_{t+1}} \right)^2}} \quad (5)$$

[Meese and Rogoff \(1983\)](#), and most subsequent studies, reached a conclusion on the superiority or otherwise of the random walk by comparing the numerical values of the RMSEs and similar metrics, without testing for the statistical significance of the difference. Some studies, however, use the [Diebold and Mariano \(1995\)](#) test. As a robustness check, we opt to use a different test—the AGS test suggested by [Ashley et al. \(1980\)](#). This test requires the estimation of the linear regression

$$D_t = \alpha_0 + \alpha_1(M_t - \bar{M}) + u_t \quad (6)$$

where  $D_t = w_{1t} - w_{2t}$ ,  $M_t = w_{1t} + w_{2t}$ ,  $\bar{M}$  is the mean of  $M$ ,  $w_{1t}$  is the forecasting error at time  $t$  of the model with the higher RMSE,  $w_{2t}$  is the forecasting error at time  $t$  of the model with the lower RMSE. If the sample mean of the errors is negative, the observations of the series must be multiplied by  $-1$  before running the regression. The estimates of the intercept term ( $\alpha_0$ ) and the slope ( $\alpha_1$ ) are used to test the statistical difference between the RMSEs of two different models. If the estimates of  $\alpha_0$  and  $\alpha_1$  are both positive, then a test of the joint hypothesis  $H_0: \alpha_0 = \alpha_1 = 0$  is appropriate. However, if one of the estimates is negative and statistically significant then the test is inconclusive. But if one of the coefficients is negative and statistically insignificant the test remains conclusive, in which case significance is determined by the upper-tail of the  $t$ -test on the positive coefficient estimate. In this study we only report the  $\chi^2(2)$  statistic for the null  $\alpha_0 = \alpha_1 = 0$ .

## 2.2. Data and results

The empirical results are based on six currency combinations involving the U.S. dollar (USD), Japanese yen (JPY), British pound (GBP) and Canadian dollar (CAD). Hence, the model is estimated for six exchange rates, three against the dollar (JPY/USD, GBP/USD and CAD/USD) and three cross rates (JPY/CAD, GBP/CAD and JPY/GBP). Monthly data covering the period January 1998–May 2013 are used. The data source is the IMF's *International Financial Statistics*. The sample period is split at December 2007 into an estimation period and a forecasting period, so that forecasts are generated over the period January 2008–May 2013. In this exercise, therefore,  $n = 185$ ,  $m = 120$  and  $k = 1, 3, 6$ .

Three considerations led to the choice of currency combinations used here. First, they involve three of the four major currencies (USD, JPY and GBP)—only the euro is excluded because of the unavailability of data on the corresponding macroeconomic variables (for the Eurozone, which keeps on changing as new countries enter the zone as members). Second, the six currency pairs account for over 30% of the global foreign exchange market turnover ([Bank for International Settlements, 2013](#)).<sup>3</sup> Third, the use of cross rates is warranted by the possibility of a potential “dollar bias” phenomenon.

[Table 1](#) reports the (basic) results of forecasting accuracy of the static monetary model (Eq. (1)) for three forecasting horizons: one month, three months and six months, including the significance of differences in RMSEs (of the model and random

<sup>3</sup> The JPY/USD, GBP/USD and CAD/USD combinations account for 18.3%, 8.8% and 3.7% of the global currency turnover, respectively.

**Table 1**

The basic results (Eq. (1)).

|                     | CAD/USD | GBP/CAD | GBP/USD | JPY/CAD | JPY/GBP | JPY/USD |
|---------------------|---------|---------|---------|---------|---------|---------|
| <i>One month</i>    |         |         |         |         |         |         |
| RMSE                | 0.107   | 0.346   | 0.389   | 0.119   | 0.317   | 0.307   |
| <i>U</i>            | 4.28    | 11.93   | 10.51   | 2.20    | 5.37    | 9.03    |
| AGS $\chi^2(2)$     | 241.77  | 4377.20 | 1874.10 | 41.57   | 1130.10 | 2389.30 |
| <i>Three months</i> |         |         |         |         |         |         |
| RMSE                | 0.136   | 0.367   | 0.408   | 0.146   | 0.322   | 0.304   |
| <i>U</i>            | 2.61    | 6.67    | 6.38    | 1.74    | 3.19    | 4.41    |
| AGS $\chi^2(2)$     | 50.17   | 429.33  | 56.98   | 30.68   | 40.58   | 440.63  |
| <i>Six months</i>   |         |         |         |         |         |         |
| RMSE                | 0.179   | 0.397   | 0.453   | 0.190   | 0.375   | 0.361   |
| <i>U</i>            | 2.27    | 7.41    | 5.15    | 1.44    | 2.57    | 3.41    |
| AGS $\chi^2(2)$     | 20.29   | 391.07  | 19.10   | 12.64   | 26.73   | 649.60  |

RMSE is the root mean square error in percentage terms. *U* is Theil's inequality coefficient. AGS is the [Ashley et al. \(1980\)](#) test whereby a significant  $\chi^2(2)$  statistic implies that the RMSEs are not equal. The 5% critical value of  $\chi^2(2)$  is 5.99.

walk) as judged by the AGS test. In all cases the random walk outperforms the monetary model in terms of the root mean square error (because  $U > 1$ ) while the AGS test rejects the null of the equality of the RMSEs. These findings support the results of [Meese and Rogoff \(1983\)](#) in the narrow sense that the model does not produce a numerically smaller RMSE than that of the random walk. In this narrow sense, therefore, the Meese–Rogoff results cannot be overturned, but this does not mean that the random walk cannot be outperformed in a broader sense that involves a consideration of direction accuracy and profitability. The next step is to demonstrate that if other measures of forecasting accuracy are used, the random walk can be outperformed.

### 3. Alternative measures of forecasting accuracy

#### 3.1. The RMSE as a measure of forecasting accuracy

The proposition that conventional macroeconomic and time series models cannot outperform the random walk in out-of-sample forecasting should be qualified if it is to reflect the available empirical evidence and status quo. If the proposition is a valid representation of the current state of affairs, it should read: “conventional macroeconomic and time series models cannot outperform the random walk in out-of-sample forecasting if forecasting accuracy is measured in terms of the root mean square error or similar metrics that depend on the magnitude of the forecasting error”. [Moosa \(2013\)](#) uses a simulation exercise to demonstrate why it is so difficult (though perhaps not impossible) to outperform the random walk in terms of the RMSE. The forecasting error of the random walk is the period-to-period change in the exchange rate, which would be small for a relatively stable exchange rate. As the exchange rate becomes increasingly unstable (hence unpredictable), the RMSE of the random walk rises, but so does the RMSE of the model (more unstable exchange rates are more difficult to forecast). The model will not outperform the random walk if the RMSE of the random walk rises more rapidly than that of the random walk as the exchange rate becomes more unstable. The results of the simulation exercise show that failure to outperform the random walk, in both in-sample and out-of-sample forecasting, should be the rule rather than the exception. It should not be taken to imply the failure of international monetary economics or represent a puzzle. Moreover, the results do not imply that the random walk is unbeatable, because it can be easily outperformed if forecasting accuracy is judged according to criteria such as direction accuracy and profitability, as we are going to see later.

#### 3.2. Direction accuracy and profitability

It has been suggested that the use of the RMSE and similar criteria to measure forecasting accuracy may not be entirely appropriate. Some economists argue that a correct prediction of the direction of change can be more important than the magnitude of the error, while others have suggested that the ultimate test of forecasting power is the ability to make profit by trading on the basis of the forecasts. However, whether the prediction of the magnitude of change is more or less important than the prediction of the direction of change depends on the underlying situation.<sup>4</sup>

[Cheung et al. \(2005\)](#) point out that using criteria other than the mean square error does not boil down to “changing the rules of the game” and that minimizing the mean square error may not be important from an economic standpoint. They present a reason for not relying on the mean square error, suggesting that it may miss out on important aspects of prediction,

<sup>4</sup> For example, in intra-day currency trading, where the interest rate differential is ignored, the prediction of the direction of change is the only thing that matters. When interest rates are taken into account, both factors become important because the decision depends on the expected rate of return, which consists of the interest rate differential and the expected change in the exchange rate. When the underlying situation involves speculation on combined currency option positions, such as a straddle, the only thing that matters is the magnitude of change.

particularly at long horizons. They also argue that the direction of change is “perhaps more important from a market timing perspective”. [Christoffersen and Diebold \(1998\)](#) contend that the mean square error indicates no improvement in predictions that take into account cointegrating relations *vis-a-vis* univariate prediction. [Leitch and Tanner \(1991\)](#) argue that the direction of change may be more relevant for profitability and economic concerns, while [Cumby and Modest \(1987\)](#) suggest that it is also related to market timing ability. [Lam et al. \(2008\)](#) find evidence indicating that exchange rate movements may be predictable at long time horizons by using direction accuracy as a measure of forecasting power.

[Engel \(1994\)](#) uses a Markov-switching model to represent 18 exchange rates at a quarterly frequency. While he finds that this model cannot outperform the random walk in terms of the magnitude of the error, he provides evidence indicating that the model is superior at predicting the direction of change. Engel, therefore, advocates the use of direction accuracy, which he describes as “not a bad proxy for a utility-based measure of forecasting performance”. He refers to the case of central banks under a fixed exchange rate system as an important example where the direction of change is exactly the right criterion for maximizing the welfare of the forecaster.

Profitability, or in general utility, may be the ultimate test of predictive power. [Abhyankar et al. \(2005\)](#) propose a utility-based criterion pertaining to the portfolio allocation problem. They find that the relative performance of a structural model improves when this criterion is used. Likewise, [West et al. \(1993\)](#) suggest a utility-based evaluation of exchange rate predictability. [Li \(2011\)](#) evaluates the effectiveness of economic fundamentals in enhancing carry trade, concluding that the profitability of carry trade and risk-return measures can be enhanced by using forecasts. Likewise, [Boothe and Glassman \(1987\)](#) compare the rankings of alternative exchange rate models using as evaluation criteria conventional measures of accuracy and profitability. The results show that the random walk ranks highest in forecasting accuracy and in terms of profitability for one of the two currency pairs used by them (German mark/U.S. dollar).

[Corte et al. \(2008\)](#) attribute failure to beat the random walk to the use of improper criteria. They assess the economic value of the in-sample and out-of-sample forecasting power of some empirical models and conclude that strategies based on combined forecasts yield large economic gains over the random walk benchmark. They also argue that the statistical evidence of exchange rate predictability does not guarantee that an investor can earn profits from an asset allocation strategy that exploits this predictability.

[Leitch and Tanner \(1991\)](#) point out that economists are puzzled by the observation that profit-maximizing firms buy professional forecasts when measures of forecasting accuracy (meaning the RMSE and similar metrics) indicate that a naïve model forecasts about as well (typically better). The reason they present is that these measures bear a very weak relation to the profit generated by acting on the basis of the forecasts and that the only substitute criterion for profit is a measure of direction accuracy. They further suggest that if profits are not observable, direction accuracy of the forecasts may be used as the evaluation criterion.

### 3.3. Methodology: direction accuracy

We use the conventional measure of direction accuracy

$$DA = \frac{1}{n - m - k + 1} \sum_{t=m+k}^n a_t \quad (7)$$

where

$$a_t = \begin{cases} 1 & \text{if } \left\{ \begin{array}{l} (\hat{S}_{t+1} - S_t)(S_{t+1} - S_t) > 0 \\ (\hat{S}_{t+1} - S_t)(S_{t+1} - S_t) < 0 \end{array} \right. \\ 0 & \end{cases} \quad (8)$$

A conventional test of the significance of proportions is used to find out if direction accuracy is significantly different from 0. A rejection of  $H_0: DA = 0$  means that the underlying model is superior to the random walk in predicting direction.

[Cheung et al. \(2005\)](#) use the higher benchmark of  $DA = 0.5$  to judge the superiority or otherwise of a model over the random walk. The rationale for using this benchmark is that the random walk “predicts the exchange rate has an equal chance to go up or down”, which sounds like a 50–50 situation. However, the random walk without drift produces no-change forecasts, since the forecast for point in time  $t$  is the actual value at  $t - 1$ . [Evans and Lyons \(2005\)](#) state explicitly that if the *ex ante* forecast follows a random walk without drift, “there is no forecast change in the spot rate”. This is because a random walk without drift is represented by  $s_t - s_{t-1} = \varepsilon_t$  or  $\Delta s_t = \varepsilon_t$ , but a property of this process is that  $E(\varepsilon_t) = 0$ . Hence for a random walk without drift  $DA = 0$ , which means that the null hypothesis should be  $H_0: DA = 0$  rather than  $H_0: DA = 0.5$ . This is rather intuitive: a model that forecasts the direction of change correctly on, say, 40% of the occasions is better than the random walk that cannot forecast the change correctly at all. It seems rather implausible to require a model to forecast direction correctly more than 50% of the times to be better than the random walk which cannot forecast direction at all.

A further test of direction accuracy is that of [Pesaran and Timmermann \(1992\)](#). The PT test is a non-parametric test for independence between the change predicted by the model and the actual change in the exchange rate. More precisely, it is a test of the significance of the difference between the observed probability of a correctly signed forecast and the estimate of the probability under the null that the forecast and actual outcomes are independent (meaning that the forecast series has no power in predicting the direction of change in the actual series). The test statistic, which is calculated from the difference

between the sample estimate of the probability of a correct signed forecast,  $\hat{P}$ , and the estimate of the expected value under the null,  $\hat{P}^*$ . Hence

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\text{var}(\hat{P}) - \text{var}(\hat{P}^*)}} \sim N(0, 1) \quad (9)$$

The test statistic has a 5% critical value of 1.64. A rejection of the null implies that the model has a significant predictive power for the direction of change.

Moosa and Burns (2012) suggest a measure of forecasting accuracy, the adjusted root mean square error (ARMSE), which combines the magnitude of the error and the ability of the model to predict direction correctly. It can be constructed by adjusting the conventional RMSE to take into account the ability or otherwise to predict the direction of change. If two models have equal RMSEs, the model with the lower DA should have a higher ARMSE. This measure of forecasting accuracy is calculated as

$$\text{ARMSE} = \sqrt{\frac{(1 - DA)}{n - m - k + 1} \sum_{t=m+k}^n \left( \frac{\hat{S}_t - S_t}{S_t} \right)^2} \quad (10)$$

where  $1 - DA$  is the confusion rate.

### 3.4. Methodology: proximity to a perfect forecast

Moosa and Burns (2013b) suggest a test of forecasting accuracy based on the following regression of the forecast change in the exchange rate on the actual change

$$\hat{S}_t - S_{t-1} = \alpha + \beta(S_t - S_{t-1}) \quad (11)$$

By imposing the restrictions  $(\alpha, \beta) = (0, 1)$  we obtain the equation for the line of perfect forecast. Any violation of the coefficient restrictions defining the line of perfect forecast implies less than perfect forecasts, invariably involving magnitude and direction errors. Either of the conditions  $\alpha \neq 0$  and  $\beta \neq 1$  may imply a combination of errors of magnitude and direction.

It follows, therefore, that a measure of forecasting accuracy that combines both magnitude and direction is the extent of deviation from the coefficient restriction  $(\alpha, \beta) = (0, 1)$ . A Wald test of coefficient restrictions can be conducted to find out if the violation is statistically significant as implied by the  $\chi^2$  statistic. If it is, a comparison can be made between a model and the random walk on the basis of the numerical value of the  $\chi^2$  statistic, such that the bigger the value, the greater the violation of the coefficient restriction and the worse is the model with respect to predictive power as judged by magnitude and direction. For the random walk to outperform the model it must produce a smaller  $\chi^2$  statistic than the model for the restriction  $(\alpha, \beta) = (0, 1)$ .

### 3.5. Methodology: Profitability

Abhyankar et al. (2005) associate the ability to predict the direction of change with “economic value”, which is a general term for profitability. While they argue that “there are many different ways to characterize or define economic value”, they follow West et al. (1993) by using a Bayesian framework to study asset allocation. Their framework for measuring profitability involves a buy-and-hold strategy and the allocation of wealth between two assets that are identical in all aspects except for the currency of denomination. Thus they consider an optimization problem to determine the optimal portfolio weights, with emphasis on the end-of-period approach. As a measure of profitability they use the wealth ratio, which is the ratio of end-of-period wealth accumulated by using fundamentals and random walk to the initial wealth.

We suggest a more intuitive approach that involves period-by-period trading by using two alternative strategies: pure carry trade and forecasting-based trading. Under the random walk (without drift), the forecast change in the exchange rate is always zero, which means that a profitable strategy would be to go short on the low-interest currency and long on the high-interest currency. This operation represents the common carry trade, which in effect is also a forecasting-based strategy except that the forecasts are generated by the random walk (without drift). Under this strategy, the period-to-period return is calculated as

$$\pi = \begin{cases} (i_b - i_a) + \dot{S}_{t+1} & \text{if } i_b > i_a \\ (i_a - i_b) - \dot{S}_{t+1} & \text{if } i_b < i_a \end{cases} \quad (12)$$

where  $i_a$  is the interest rate on currency  $a$ ,  $i_b$  is the interest rate on currency  $b$  and  $\dot{S}_{t+1}$  is the percentage change in the exchange rate. On the other hand, if forecasts are used for trading, the decision rule will be based on whether the forecast return,  $\hat{\pi}$ , is positive or negative. In this case the realized return is calculated as

$$\pi = \begin{cases} (i_b - i_a) + \dot{S}_{t+1} & \text{if } \hat{\pi} > 0 \\ (i_a - i_b) - \dot{S}_{t+1} & \text{if } \hat{\pi} < 0 \end{cases} \quad (13)$$

**Table 2**

Measures of forecasting accuracy of the random walk.

|                     | CAD/USD   | GBP/CAD | GBP/USD   | JPY/CAD   | JPY/GBP | JPY/USD |
|---------------------|-----------|---------|-----------|-----------|---------|---------|
| <i>One month</i>    |           |         |           |           |         |         |
| RMSE                | 0.025     | 0.029   | 0.037     | 0.054     | 0.059   | 0.034   |
| ARMSE               | 0.025     | 0.029   | 0.037     | 0.054     | 0.059   | 0.034   |
| Wald $\chi^2(2)$    | 162071.90 | 1947.70 | 100482.60 | 113691.10 | 4828.60 | 1124.00 |
| $\bar{\pi}$         | −0.67     | 4.77    | −4.78     | 1.39      | −6.10   | −6.46   |
| SR                  | −0.02     | 0.15    | −0.13     | 0.02      | −0.14   | −0.17   |
| <i>Three months</i> |           |         |           |           |         |         |
| RMSE                | 0.052     | 0.055   | 0.064     | 0.084     | 0.101   | 0.069   |
| ARMSE               | 0.052     | 0.055   | 0.064     | 0.084     | 0.101   | 0.069   |
| Wald $\chi^2(2)$    | 27807.00  | 1184.80 | 5803488.0 | 74382.40  | 4510.90 | 684.45  |
| $\bar{\pi}$         | −0.43     | 3.47    | −5.04     | 0.41      | −4.31   | −7.47   |
| SR                  | −0.02     | 0.23    | −0.03     | 0.01      | −0.12   | −0.33   |
| <i>Six months</i>   |           |         |           |           |         |         |
| RMSE                | 0.079     | 0.058   | 0.092     | 0.132     | 0.148   | 0.106   |
| ARMSE               | 0.079     | 0.058   | 0.092     | 0.132     | 0.148   | 0.106   |
| Wald $\chi^2(2)$    | 34044.40  | 1129.40 | 873269.30 | 36818.90  | 2907.90 | 585.61  |
| $\bar{\pi}$         | −0.30     | 3.99    | −5.23     | 0.21      | −4.36   | −10.13  |
| SR                  | −0.02     | 0.38    | −0.25     | 0.01      | −0.18   | −0.31   |

ARMSE is the adjusted root mean square error. Mean returns and the Sharpe ratio are denoted  $\bar{\pi}$  and SR, respectively. Wald is a test for the proximity to a perfect forecast, such that a significant test statistic indicates deviation from a perfect forecast.

where  $\hat{\pi} = (i_b - i_a) + \hat{S}_{t+1}$  and  $\hat{S}_{t+1}$  is the forecast percentage change in the exchange rate.<sup>5</sup> Profitability is assessed in terms of the mean return, the standard deviation and the Sharpe ratio. For a sample size  $n - m - k + 1$ , where  $t = m + k, \dots, n$ , the mean and standard deviation are calculated as

$$\bar{\pi} = \frac{1}{n - m - k + 1} \sum_{t=m+k}^n \pi_t \quad (14)$$

$$\sigma = \sqrt{\frac{1}{n - m - k} \sum_{t=m+k}^n (\pi_t - \bar{\pi})^2} \quad (15)$$

The Sharpe ratio is used to measure risk-adjusted return. Following [Burnside et al. \(2010\)](#) and [Gynelberg and Remolona \(2007\)](#), the Sharpe ratio is calculated as the ratio of the mean to the standard deviation of the rate of return:

$$SR = \frac{\bar{\pi}}{\sigma} \quad (16)$$

A model producing a higher Sharpe ratio than that of the random walk implies superior risk-adjusted profitability. In essence this approach is based on the proposition put forward by [Li \(2011\)](#) that forecasts boost the profitability of carry trade.

### 3.6. Results

[Table 2](#) reports the results for the random walk over three forecasting horizons. Since the random walk without drift predicts a zero-change in the exchange rate, it has a direction accuracy of zero, hence a confusion rate of 1, which makes the RMSE and ARMSE equal. The table also reports the results of the Wald test for proximity to a perfect forecast, as well as the mean return and the Sharpe ratio. These statistics are used for comparison with the corresponding statistics for the monetary model.

In [Table 3](#) we report measures of forecasting accuracy (specifically, measures of direction accuracy and profitability) for the flexible-price monetary model in a static form (Eq. (1)). In terms of direction accuracy, the monetary model is superior to the random walk in all cases because the null hypothesis  $H_0: DA = 0$  is rejected for all exchange rates and forecasting horizons, which means that the model overwhelmingly outperforms the random walk in terms of direction accuracy. The model actually passes the PT test in one case for the one-month horizon (GBP/USD) and in four cases for the three-month and six-month horizons. However, failure to pass the PT test does not mean that the model is as bad as the random walk in terms of direction accuracy. The model is better as long as the null  $DA = 0$  is rejected, which is so in all cases. The random walk is beaten decisively in terms of direction accuracy.

<sup>5</sup> Calculation of the rate of return as in Eqs. (12) and (13) requires adjustment to take into account transaction costs. However, the same number of trades are executed under both strategies (which is the same as the number of forecasts). This means that the transaction costs incurred under the two strategies are equal, in which case this factor is ignored.

**Table 3**

Measures of forecasting accuracy of the static model (Eq. (1)).

|                     | CAD/USD | GBP/CAD | GBP/USD | JPY/CAD | JPY/GBP | JPY/USD |
|---------------------|---------|---------|---------|---------|---------|---------|
| <i>One month</i>    |         |         |         |         |         |         |
| DA                  | 0.46    | 0.43    | 0.48    | 0.54    | 0.55    | 0.43    |
| $z(DA = 0)$         | 7.46    | 7.01    | 7.70    | 8.71    | 8.98    | 7.01    |
| PT                  | −0.06   | 0.00    | 2.46    | 1.25    | 1.10    | 0.00    |
| ARMSE               | 0.079   | 0.261   | 0.281   | 0.081   | 0.213   | 0.232   |
| Wald $\chi^2(2)$    | 41.98   | 284.56  | 226.07  | 56.95   | 199.40  | 327.11  |
| $\pi$               | 8.85    | −2.51   | −4.55   | 14.63   | −5.03   | −1.45   |
| SR                  | 0.23    | −0.08   | −0.13   | 0.27    | −0.09   | −0.04   |
| <i>Three months</i> |         |         |         |         |         |         |
| DA                  | 0.52    | 0.43    | 0.56    | 0.67    | 0.52    | 0.41    |
| $z(DA = 0)$         | 8.32    | 6.87    | 8.87    | 11.22   | 8.32    | 6.65    |
| PT                  | 1.96    | 0.20    | 2.71    | 2.78    | 0.83    | 1.94    |
| ARMSE               | 0.094   | 0.277   | 0.271   | 0.084   | 0.223   | 0.234   |
| Wald $\chi^2(2)$    | 8.32    | 37.10   | 76.47   | 31.88   | 97.85   | 294.45  |
| $\pi$               | 6.98    | −1.50   | −0.81   | 9.04    | 12.33   | −4.35   |
| SR                  | 0.35    | −0.10   | −0.03   | 0.28    | 0.12    | −0.18   |
| <i>Six months</i>   |         |         |         |         |         |         |
| DA                  | 0.65    | 0.43    | 0.72    | 0.70    | 0.53    | 0.38    |
| $z(DA = 0)$         | 10.56   | 6.77    | 12.32   | 11.83   | 8.28    | 6.11    |
| PT                  | 3.13    | −0.19   | 4.05    | 3.21    | 2.03    | 0.00    |
| ARMSE               | 0.217   | 0.300   | 0.240   | 0.104   | 0.257   | 0.284   |
| Wald $\chi^2(2)$    | 0.106   | 0.300   | 0.240   | 0.104   | 0.257   | 0.284   |
| $\pi$               | 7.30    | −1.66   | 12.91   | 8.40    | 5.79    | −3.12   |
| SR                  | 0.48    | −0.15   | 0.10    | 0.37    | 0.24    | −0.09   |

DA is direction accuracy.  $z$  is the test statistic for testing a proportion, which has a critical value of 1.96. The  $P$ -test statistic is distributed as  $N(0, 1)$  and has a critical value of 1.64 at the 5% level of significance.

In terms of the ARMSE the random walk still outperforms the model but not by such a big margin as when comparison is based on the RMSE. The explanation for this finding is simple: the random walk is so superior in terms of the RMSE that it would take very high direction accuracy for the model to outperform the random walk in terms of the ARMSE.<sup>6</sup> However, in terms of proximity to a perfect forecast (the Wald test), which also takes into account direction and magnitude, the model outperforms the random walk for all exchange rates and all horizons. In terms of profitability, the model fails to outperform the random walk in one case only, GBP/CAD (for all three horizons). Hence the random walk is beaten overwhelmingly in terms of profitability.

The static monetary model, therefore, outperforms the random walk in terms of direction accuracy and profitability. The forecasts produced by the model are much closer to the line of perfect forecast than those of the random walk. In this broader sense the findings of Meese and Rogoff are overturned. Yes the model cannot outperform the random walk in terms of the RMSE (which should not be surprising) but it produces superior forecasts when evaluated in terms of criteria other than the magnitude of the forecasting error. The model produces better forecasts in terms of profitability, which (from a business perspective at least) matters more than the RMSE. This is an explanation for the observation why profit-maximizing firms buy professional forecasts although they are inferior to those generated by the random walk in terms of the RMSE. This observation does not represent a puzzle at all.

#### 4. The use of dynamic specifications

##### 4.1. Methodology

Some economists have attempted to boost the forecasting power of exchange rate models by introducing dynamics, including the use of error correction mechanisms. For example, Taylor (1995) suggests that “researchers have found that one key to improving forecast performance based on economic fundamentals lies in the introduction of equation dynamics”. Taylor points out that “this has been done in various ways: by using dynamic forecasting equations for the forcing variables in the forward-looking, rational expectations version of the flexible-price monetary model, by incorporating dynamic partial adjustment terms into the estimating equation, by using time-varying parameter estimation techniques, and—most recently—by using dynamic error correction forms”.

Claims of “victory” over the random walk by introducing dynamics have been made by Cheung et al. (2005), Tawadros (2001), Chinn and Meese (1995), Hwang (2001), and Aarle et al. (2000). Some of these claims, however, are made without

<sup>6</sup> For example, in the case of the CAD/USD rate at the one-month horizon, direction accuracy must be 0.95 in order for the model to have the same ARMSE as the random walk.

**Table 4**

Measures of forecasting accuracy of the error correction model (Eq. (17)).

|                     | CAD/USD | GBP/CAD | GBP/USD | JPY/CAD | JPY/GBP | JPY/USD |
|---------------------|---------|---------|---------|---------|---------|---------|
| <i>One month</i>    |         |         |         |         |         |         |
| RMSE                | 0.027   | 0.030   | 0.038   | 0.058   | 0.046   | 0.022   |
| <i>U</i>            | 1.08    | 1.03    | 1.03    | 1.07    | 0.78    | 0.65    |
| AGS $\chi^2(2)$     | 31.30   | 52.32   | 29.86   | 12.15   | 14.48   | 10.48   |
| <i>DA</i>           | 0.52    | 0.45    | 0.42    | 0.52    | 0.58    | 0.55    |
| <i>z(DA = 0)</i>    | 8.44    | 7.24    | 6.80    | 8.44    | 9.56    | 8.98    |
| <i>PT</i>           | 0.62    | −0.54   | 0.53    | 0.20    | 1.35    | 1.33    |
| ARMSE               | 0.109   | 0.022   | 0.029   | 0.040   | 0.030   | 0.015   |
| Wald $\chi^2(2)$    | 575.91  | 121.19  | 617.40  | 1910.60 | 641.80  | 2044.90 |
| $\pi$               | −4.17   | −0.73   | −4.78   | 7.36    | 4.79    | −2.48   |
| <i>SR</i>           | −0.11   | −0.02   | −0.13   | 0.13    | 0.09    | −0.07   |
| <i>Three months</i> |         |         |         |         |         |         |
| RMSE                | 0.069   | 0.071   | 0.078   | 0.095   | 0.085   | 0.064   |
| <i>U</i>            | 1.32    | 1.23    | 1.22    | 1.13    | 0.84    | 0.93    |
| AGS $\chi^2(2)$     | 38.26   | 62.05   | 35.38   | 12.75   | 15.56   | 15.13   |
| <i>Six months</i>   |         |         |         |         |         |         |
| RMSE                | 0.107   | 0.109   | 0.117   | 0.135   | 0.124   | 0.102   |
| <i>U</i>            | 1.35    | 1.89    | 1.27    | 1.02    | 0.85    | 0.96    |
| AGS $\chi^2(2)$     | 39.14   | 100.61  | 35.38   | 11.56   | 7.21    | 8.32    |

For notation, see footnotes to previous tables.

appropriate testing for the significance of the difference in RMSEs. For example, Hwang (2001) finds the RMSEs (measured in percentage terms) of the Frenkel–Bilson model, the Dornbusch–Frankel model and the random walk to be 1.139, 1.138 and 1.174, respectively. According to these figures, he concludes that both models outperform the random walk. It is rather misleading to conclude that the two models are superior to the random walk when the numerical differences between the RMSEs are so small that they are unlikely to be statistically significant.

A more serious observation is that the use of dynamics, including error correction models, implicitly and effectively boils down to the introduction of a lagged dependent variable, which makes the underlying model some sort of an augmented random walk. The random walk component, which is represented by the lagged dependent variable, typically dominates the effect of the explanatory variables suggested by theory to be important determinants of the exchange rate. Schinasi and Swamy (1989) view a structural model with a lagged dependent variable as a model in which the lagged variable (representing a random walk process) and the explanatory variables “are allowed to explain the spot exchange rate”. However, experience shows that the random walk component invariably prevails (Kling, 2010, 2011), in which case a random walk with explanatory variables is unlikely to perform better (in terms of the RMSE) than a pure random walk. And even if it does, beating the random walk with another random walk does not sound exactly right.

It may be disingenuous, therefore, to claim that a model outperforms the random walk, just because it has been augmented by a random walk component (even worse if the claim is made without appropriate testing). Not every study using dynamic models, however, produces results indicating the superiority of the model over the random walk. For example, Fullerton et al. (2001) used a set of error correction models to represent the behavior of the exchange rate between the Mexican peso and the U.S. dollar. Their results led them to conclude that “although dynamic simulation properties of the equations are acceptable, in no case do they generate levels of accuracy that exceed that associated with a simple random walk”.<sup>7</sup>

Moosa and Burns (2013c) examine the proposition that improvement in the forecasting accuracy of a dynamic model relative its static version is due to the introduction of a random walk component represented by a lagged dependent variable. They estimate four dynamic versions of the flexible-price monetary model for the yen/dollar exchange rate over a period extending back to 1990. Their results show that while all dynamic models outperform the static model, none of them outperforms the random walk in terms of the RMSE. To rationalize the empirical results, they show (through appropriate algebraic manipulation) that all of the dynamic models can be re-specified in levels with lagged dependent variables. This representation holds for any dynamic specification, including those based on partial adjustment and distributed lags in the dependent and explanatory variables, as well as straight first difference and error correction models.

#### 4.2. Results

The results presented in Table 4 are based on forecasts generated from the error correction model corresponding to Eq. (1).<sup>8</sup> The model is specified as

<sup>7</sup> Failure to exceed the accuracy of the random walk implies the dominance of the random walk component as represented by a lagged dependent variable.

<sup>8</sup> Results of direction accuracy and profitability are presented only for the one-month horizon because for other horizons they are qualitatively similar. These results are available from the authors upon request.

$$\Delta s_t = \alpha_0 + \sum_{j=1}^{\ell} \alpha_j \Delta s_{t-j} + \sum_{j=0}^{\ell} \beta_j \Delta(m_{t-j} - m_{t-j}^*) + \sum_{j=0}^{\ell} \gamma_j \Delta(y_{t-j} - y_{t-j}^*) + \sum_{j=0}^{\ell} \delta_j \Delta(i_{t-j} - i_{t-j}^*) + \phi \varepsilon_{t-1} + \xi_t \quad (17)$$

As we can see, the forecasting power of the model improves dramatically in terms of the RMSE. For two of the three exchange rates involving the Japanese yen, the RMSE of the model is significantly lower than that of the random walk (for all horizons). This may be taken to mean that the random walk can indeed be outperformed in terms of the RMSE, which overturns the Meese–Rogoff contention even in a narrow sense. However, the significant improvement in the performance of the model (in terms of the RMSE) can be attributed to the introduction of a random walk term represented by the lagged dependent variable. This proposition can be demonstrated by simplifying the error correction model (Eq. (17)) such that the three explanatory variables are replaced with a vector,  $x_t$ , while imposing the restriction  $\ell = 1$ . Since  $\Delta s_t = s_t - s_{t-1}$  and  $\Delta x_t = x_t - x_{t-1}$ , Eq. (17) becomes

$$s_t - s_{t-1} = \alpha_0 + \alpha_1(s_{t-1} - s_{t-2}) + \beta_0(x_t - x_{t-1}) + \beta_1(x_{t-1} - x_{t-2}) + \phi(s_{t-1} - a_0 - a_1 x_{t-1}) + \xi_t \quad (18)$$

which gives

$$s_t = (\alpha_0 - \phi a_0) + (1 + \alpha_1 + \phi)s_{t-1} - \alpha_1 s_{t-2} + \beta_0 x_t + (-\beta_0 + \beta_1 - \phi a_1)x_{t-1} - \beta_1 x_{t-2} + \xi_t \quad (19)$$

The process  $s_t = (\alpha_0 - \phi a_0) + (1 + \alpha_1 + \phi)s_{t-1} + \xi_t$  represents random walk without drift if  $(\alpha_0 - \phi a_0) = 0$  and  $(1 + \alpha_1 + \phi) = 1$ . Because the exchange rate is an integrated process, the value of the coefficient on the lagged dependent variable is typically close to one (insignificantly different from one). Furthermore, we recall [Kling's \(2010\)](#) assertion that the coefficient on the lagged dependent variable tends to be close to one, which he attributes to time aggregation. Hence, the finding of Meese and Rogoff that the random walk cannot be outperformed in terms of the RMSE still stands.<sup>9</sup>

One point that needs clarification here is why the introduction of dynamics produces significant improvement over the static model in terms of the RMSE, but not in terms of the DA. While the static model passes the PT test in one case (for the one-month horizon), the dynamic model does not at all. The reason for this disparity is that the dynamic model is overwhelmed by the random walk component. As we know by now, the random walk performs very well in terms of the RMSE while it is no good at all in terms of the ability to predict the direction of change. It remains the case, however, that the dynamic model outperforms the random walk in terms of direction accuracy because the null  $DA = 0$  is rejected in all cases. It also produces forecasts that are closer to the line of perfect forecast than those of the random walk.

## 5. The use of TVP estimation

### 5.1. The rationale

[Schinasi and Swamy \(1989\)](#) advocate the use of TVP estimation of exchange rate models for a number of reasons. The first is that model parameters can change over time because traders do not use information in the same way over all policy regimes and all time horizons. The second is that, because of the heterogeneity of market participants, macroeconomic variables are not related to the exchange rate by a simple fixed-coefficient relationship. The third is that the use of fixed coefficients implies the imposition of a restriction that may or may not be valid. For example, [Moosa and Kwiecien \(2002\)](#) argue that the nominal interest rate is more capable of predicting inflation if the assumption of fixed coefficients is relaxed. The implications of representing the Fisher equation by a fixed coefficient regression equation are that the real interest rate is fixed and that the response of the nominal interest rate to inflationary expectations does not change over time. These assumptions are implausible.

[Schinasi and Swamy \(1989\)](#) estimate models with stochastic coefficients that encompass as special cases the Kalman filtering technique, the method of [Hildreth and Houk \(1968\)](#) and ARCH models. While the results obtained by using fixed-coefficient models support the Meese–Rogoff conclusion, they find that the random walk is outperformed by the TVP versions of the flexible-price model, the sticky-price model and the sticky-price model with current accounts (the three models used by Meese and Rogoff). The problem is that they base their conclusions on the numerical values of the mean absolute error and root mean square error. We propose a robustness check on these results by using a different TVP estimation method and by subjecting the estimated RMSEs to the AGS test.

### 5.2. Methodology

The procedure used in this paper for TVP estimation follows the structural time series approach ([Harvey, 1989; Koopman et al., 2006](#)). For the monetary model to be estimated in a TVP framework we specify it as

$$s_t = \mu_t + \phi_t + \alpha_{1t}(m_{a,t} - m_{b,t}) + \alpha_{2t}(y_{a,t} - y_{b,t}) + \alpha_{3t}(i_{a,t} - i_{b,t}) + \varepsilon_t \quad (20)$$

where  $\mu_t$ ,  $\phi_t$  and  $\varepsilon_t$  are the (unobserved) time series components of  $s_t$ :  $\mu_t$  is the trend,  $\phi_t$  is the cyclical component and  $\varepsilon_t$  is the random component. The trend, which represents the long-term movement of the dependent variable, is represented by the general specification

<sup>9</sup> We also tried a nonlinear version of Eq. (17) by including a polynomial of degree three in the error correction term. We found that the improvement in the forecasting power of the model was due to the introduction of dynamics rather than nonlinearity.

**Table 5**

Measures of forecasting accuracy of the TVP model (Eq. (20)).

|                     | CAD/USD | GBP/CAD | GBP/USD | JPY/CAD | JPY/GBP | JPY/USD |
|---------------------|---------|---------|---------|---------|---------|---------|
| <i>One month</i>    |         |         |         |         |         |         |
| RMSE                | 0.047   | 0.037   | 0.046   | 0.044   | 0.044   | 0.022   |
| <i>U</i>            | 1.23    | 1.43    | 1.30    | 0.81    | 0.85    | 0.95    |
| AGS $\chi^2(2)$     | 12.64   | 19.36   | 17.69   | 4.64    | 14.40   | 14.60   |
| <i>DA</i>           | 0.63    | 0.72    | 0.43    | 0.78    | 0.85    | 0.66    |
| $z(DA = 0)$         | 10.54   | 13.03   | 7.01    | 15.39   | 18.91   | 11.27   |
| <i>PT</i>           | 2.68    | 3.77    | −0.06   | 4.60    | 5.61    | 2.57    |
| ARMSE               | 0.027   | 0.020   | 0.035   | 0.021   | 0.017   | 0.013   |
| Wald $\chi^2(2)$    | 4.98    | 12.63   | 44.57   | 3.08    | 21.66   | 94.04   |
| $\pi$               | 13.12   | 19.16   | −2.00   | 34.37   | 36.59   | 12.37   |
| <i>SR</i>           | 0.35    | 0.72    | −0.06   | 0.77    | 0.94    | 0.35    |
| <i>Three months</i> |         |         |         |         |         |         |
| RMSE                | 0.083   | 0.077   | 0.085   | 0.083   | 0.086   | 0.064   |
| <i>U</i>            | 1.59    | 1.33    | 1.32    | 0.99    | 0.82    | 0.93    |
| AGS $\chi^2(2)$     | 16.39   | 8.70    | 7.82    | 4.31    | 9.08    | 8.52    |
| <i>Six months</i>   |         |         |         |         |         |         |
| RMSE                | 0.122   | 0.117   | 0.124   | 0.122   | 0.127   | 0.102   |
| <i>U</i>            | 1.55    | 2.02    | 1.35    | 0.93    | 0.86    | 0.96    |
| AGS $\chi^2(2)$     | 14.87   | 19.97   | 18.77   | 4.94    | 12.62   | 11.40   |

For notation, see footnotes to previous tables.

$$\mu_t = \mu_{t-1} + \beta_{t-1} + \eta_t \quad (21)$$

$$\beta_t = \beta_{t-1} + \zeta_t \quad (22)$$

where  $\eta_t \sim NID(0, \sigma_\eta^2)$ , and  $\zeta_t \sim NID(0, \sigma_\zeta^2)$ . The cyclical component is specified as

$$\phi_t = \rho(\phi_{t-1} \cos \theta + \phi_{t-1}^* \sin \theta) + \omega_t \quad (23)$$

$$\phi_t^* = \rho(-\phi_{t-1} \sin \theta + \phi_{t-1}^* \cos \theta) + \omega_t^* \quad (24)$$

where  $\phi_t^*$  appears by construction such that  $\omega_t$  and  $\omega_t^*$  are uncorrelated white noise disturbances with variances  $\sigma_\omega^2$  and  $\sigma_{\omega^*}^2$ , respectively. The parameters  $0 \leq \theta \leq \pi$  and  $0 \leq \rho \leq 1$  are the frequency of the cycle and the damping factor on the amplitude, respectively. The period of the cycle, which is the time taken by the cycle to go through its complete sequence of values, is  $2\pi/\theta$  (Harvey, 1989; Koopman et al., 2006).

The model is estimated in a TVP framework using maximum likelihood and the Kalman filter to update the state vector as described by Koopman et al. (1999, 2006) and Moosa (2006). Koopman et al. (2006) point out that the statistical treatment of the structural time series model (such as that represented by Eq. (20)) is based on the state-space form. Koopman et al. (1999) demonstrate how the model is represented in state-space form. Moosa (2006) presents the specification of the measurement and transition equations for structural time series models, which form the state-space representation required for TVP estimation by the Kalman filter.<sup>10</sup> The use of unobserved components is motivated by the desire to obtain a general model in which the explanatory variables that do not appear explicitly on the right-hand side of the equation are accounted for by the trend and cycle. If these components are statistically significant, this means that some unidentified variables affect the exchange rate. In this sense the model represented by Eq. (20) is more general than the three models estimated by Meese and Rogoff (1983).<sup>11</sup>

### 5.3. Results

The results of forecasting accuracy of the TVP model represented by Eq. (20) are exhibited in Table 5. The TVP model produces better results in terms of the RMSE than the static model, and for two exchange rates (JPY/GBP and JPY/USD) the results are even better than those of the random walk as judged by the AGS test. To make the same common mistake found in the literature, we could jump to the flawed conclusion that the monetary model in a TVP form outperforms the random walk in terms of the RMSE for the JPY/CAD rate because *U* is less than one. In this case, however, the difference between the root mean square errors is insignificant as indicated by the AGS test—hence the model is not superior to the random walk in terms of the RMSE for this particular exchange rate.

<sup>10</sup> TVP estimation is carried out by using the STAMP software (Structural Time Series Analyser, Modeller and Predictor). See, for example, Mendelsohn (2011) who describes STAMP as combining “state-space methods with unobserved components”, suggesting that it can be used to estimate a wide variety of both univariate and multivariate state-space models.

<sup>11</sup> This is indeed another reason why the use of one model (the flexible-price monetary model) is sufficient for the purpose of this paper.

In essence the TVP version of the monetary model produces similar results to those of the error correction model. In the case of the error correction model we attribute improvement relative to the static model (in terms of the RMSE) to the introduction of a random walk component represented by the lagged dependent variable. The question therefore is why the TVP model produces similar results as if it includes a random walk component. The answer is that this is actually the case because this particular TVP model does contain a random walk component resulting from the specification of the level and slope in Eqs. (21) and (22), respectively.  $\mu_t$  is a random walk with a drift factor,  $\beta_t$ , which follows a first order autoregressive process as represented by Eq. (22). This process collapses to a simple random walk with drift if  $\sigma_\epsilon^2 = 0$ , and to a deterministic linear trend if  $\sigma_\eta^2 = 0$  as well.<sup>12</sup> The cycle is also specified as  $\phi_t = f(\phi_{t-1})$ . Furthermore, the estimation of the TVP coefficients on the explanatory variables ( $\alpha_{1t}$ ,  $\alpha_{2t}$  and  $\alpha_{3t}$ ) involves a random walk specification. In general terms, a TVP model implicitly involves dynamics because the coefficients are allowed to change over time, which is the essence of dynamics. This is why Taylor (1995) considers TVP estimation as one way to introduce dynamics.

Again, this does not mean that the random walk cannot be outperformed—that is possible when criteria other than the RMSE are used. In terms of direction accuracy, the hypothesis  $DA = 0$  is rejected, implying that the model outperforms the random walk in all cases. There is indeed significant improvement in the ability of the model to predict the direction of change, as the TVP model passes the PT test in five out of six cases. In one case (JPY/GBP) the model predicts direction correctly on 85% of all occasions. An intuitive explanation for this result is that a change in direction is more easily picked up by time-varying than fixed coefficients.

The TVP model outperforms the random walk in terms of the ARMSE.<sup>13</sup> In terms of measures of profitability, the model outperforms the random walk by a wide margin, producing mean returns that are higher than those obtained by using straight carry trade. The results are overwhelmingly in favor of the TVP version of the monetary model, implying again that the random walk can be beaten.

## 6. Concluding remarks

Since the publication of the Meese and Rogoff (1983) paper, failure to outperform the random walk in out-of-sample forecasting has become some sort of an undisputed fact of life. In this paper we present results and arguments to suggest that the unbeatable random walk is a myth, not reality. Typically, the random walk cannot be beaten if the forecasting power of conventional macroeconomic models is judged by measures of forecasting accuracy such as the mean absolute error, mean square error and root mean square error. It is argued that this observation should be expected rather than considered a puzzle that implies the failure of international monetary economics at large. Claims of victory over the random walk (in terms of the RMSE) are invariably based on a comparison of the numerical values of measures of forecasting accuracy without proper testing of statistical significance. Victory is typically achieved by introducing a random walk component in the model as represented by a lagged dependent variable in a dynamic model. We even demonstrate that the unobserved components model, which is estimated in a TVP framework, appears to be more successful than the static model because by construction it includes dynamics and hence a random walk component. Without the use of dynamics, which is not a legitimate procedure for this purpose, it is unlikely that any model can outperform the random walk in terms of the RMSE. However, when forecasting power is measured in terms of direction accuracy and profitability, even the static model outperforms the random walk.

A question that arises here is the following: how can we explain the failure of macroeconomic models in terms of the RMSE while they are capable of outperforming the random walk in terms of direction accuracy and profitability? The answer is simple. Macroeconomic models produce significant forecasting errors because they cannot explain the stylized facts about movements in exchange rates such as bubbles followed by crashes and volatility clustering (see, for example, Moosa and Bhatti, 2010). On the other hand they can outperform the random walk in terms of direction accuracy because the random walk without drift is a no-change model. The finding that the monetary model can predict the direction of change must indicate some value in using macroeconomic fundamentals. And since profitability is related more to direction than magnitude, macroeconomic models outperform the random walk in terms of profitability.

It is rather difficult, though perhaps not impossible, for conventional macroeconomic models to outperform the random walk model in terms of the root mean square error and similar quantitative measures of forecasting accuracy that depend entirely on the absolute forecasting error. However, this failure does not represent a puzzle and should not be viewed as undermining the state of international monetary economics. If the discipline is in a bad shape, it is not because exchange rate models produce higher RMSEs than that of the random walk. The unbeatable random walk is a myth, the failure of models to outperform the random walk in terms of the RMSE is reality, and the Meese–Rogoff puzzle is anything but a puzzle.

## Acknowledgements

We are grateful to the editor of this journal and two anonymous referees for useful and detailed comments.

<sup>12</sup> Harvey (1989, pp. 510–511) lists all of the possibilities. Koopman et al. (2006) identify several models, including: a constant term model, local level model, random walk with and without drift, local level with fixed slope, a smooth trend model, etc.

<sup>13</sup> A test for the statistical significance of the difference between two ARMSEs is not available. However, it is safe to conclude that the difference is statistically significant if the difference between the corresponding RMSEs is insignificant while the difference in DAs is significant.

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